Ratan's emphasis on splitting water and the use of hydrogen has influenced the Tata companies to accord this goal a priority. It began while he was still at the helm. Tridibesh Mukherjee, Tata Steel's former chief metallurgist and director, scientific services, who is also amongst the leading metallurgists in the world, characterized Ratan as a 'scientist' in an article in 2013. He cited an example of how Ratan's scientific temper resulted in TISCO inducting a new energy source. He nudged him to 'harvest' hydrogen for use as a fuel. It resulted in Tata Steel innovating to use it in the ferrochrome plant, splitting water at high temperatures, wrote Mukherjee.

Though the shift to hydrogen as a fuel has not happened the way Ratan had envisaged or at the speed at which the transition would happen in the industry in general and particularly in automobile sector, there has been some progress in the use of this gas. While Ratan was focused on both the production and use of hydrogen, progress is being increasingly reported on finding ways to use it. Nevertheless, it goes towards the fulfilment of his emphasis on the use of alternate energy for environment protection.

The need to use hydrogen or other alternate fuel attains critical importance in India as it has the highest 'coal-based steel industry' in the world and it is the 'most carbon-intensive fossil fuel'. Working towards this goal, Tata Steel, in April 2023, for instance, began, for the first time, an 'experiment injecting hydrogen gas' in a blast furnace in its steel plant in Jamshedpur. Tata Steel MD and CEO T.V. Narendran remarked at a convention of All India Management Association (AIMA) that 'Availability of hydrogen as a fuel and reductive is the key to reducing dependence on coal' for steel making.

Though it was not what Ratan had desired (use of hydrogen in internal combustion engines) in 2023, Tata Motors delivered to Indian Oil Corporation Ltd (IOCL) two out of the fifteen hydrogen fuel cell powered (FCEV) buses to the corporation.

Tata Cummins Private Limited (TCPL), 'a joint venture of Tata Motors Limited and Cummins Inc. set up by Ratan (in 1993)', in March 2024, inaugurated a factory in Jamshedpur to produce 'hydrogen-based internal combustion engines for medium and heavy commercial vehicles'. This is more aligned to the goals that Ratan had outlined.

Girish Wagh said that 'hydrogen-based powertrain aggregates and systems [will be] produced' in the factory shaping the 'future of mobility in